

Problem Statement

Despite advances in endoscopic technology, **missed gastric lesions remain a significant clinical challenge**, with studies showing 11.3% of upper gastrointestinal (UGI) neoplasms overlooked during initial endoscopy^[1]. Missed lesions often occur due to:

- **Anatomic blind spots** (e.g., lesser curvature of the antrum, proximal duodenum)^{[2] [3]}.
- **Human factors** like rapid scope withdrawal or incomplete visualization^{[1] [4]}.
- **Cognitive biases** leading to under-recognition of subtle pathologies^{[5] [6]}.

A **real-time AI-assisted system** that guides endoscopists to systematically document all gastric regions and flags missed areas could reduce diagnostic errors, standardize examinations, and mitigate biases.

Why Photo/Video Documentation is Critical

1. Quality Control & Legal Protection

- Systematic documentation (e.g., SSS protocol) ensures completeness, with guidelines recommending **10 standardized images in upper endoscopy** (proximal esophagus to duodenum)^{[7] [3]}.
- Incomplete exams are a leading cause of malpractice claims; documentation validates procedural integrity^{[1] [4]}.

2. Reducing Missed Lesions

- Retrospective studies show **5.9–22.46% of blind spots** in conventional endoscopy vs. **2.8–5.5% with AI-guided systems**^{[3] [4]}.
- Example: Flat early gastric cancers (0-IIb type) are missed in 3.9% of cases due to incomplete imaging^[2].

3. Longitudinal Monitoring

- Serial images enable tracking of precancerous changes (e.g., intestinal metaplasia, *H. pylori* gastritis)^{[7] [8]}.

Evidence of Regional Misses in Endoscopy

Region	Miss Rate	Cause	Source
Lesser curvature (antrum)	40%	Scope retraction without upward angulation	[2] [3]
Proximal duodenum	50%	Incomplete scope advancement	[1] [3]
Gastric cardia	15%	Poor retroflexion technique	[2] [4]

Region	Miss Rate	Cause	Source
Posterior antrum	22.46%	Mucus obstruction, rapid withdrawal	[4]

How Your Model Addresses These Gaps

1. Real-Time Anatomic Landmark Detection

- AI trained to recognize **26 gastric zones** (e.g., pylorus, Z-line) with >90% accuracy^[3]^[4], ensuring all SSS protocol regions are captured.
- Example: Alerts for missed documentation of the duodenal bulb or gastric angulus.

2. Bias Mitigation

- Reduces **gender/racial disparities** in lesion detection by standardizing image acquisition^[5] ^[6].
- Flags subtle lesions (e.g., early gastric cancer) that inexperienced endoscopists might overlook^[8] ^[9].

3. Enhanced Disease Localization

- Integrates with CADe systems (sensitivity: 92.2% for EGC)^[8] to highlight suspicious areas during live exams.

Research Backing for Your Approach

1. AI Reduces Blind Spots

- Trials show AI-assisted EGD lowers blind spot rates from **22.46% to 5.86%**^[4].
- Systems like WISENSE improve photodocumentation completeness from 79.14% to 92.91%^[4].

2. Bias Reduction

- AI standardizes decision-making, countering cognitive biases (e.g., confirmation bias) in lesion assessment^[5] ^[10] ^[6].
- Models trained on diverse datasets reduce underdiagnosis in marginalized populations^[6].

3. Regulatory Alignment

- FDA-cleared AI tools (e.g., GI Genius) validate the clinical utility of real-time assistive systems^[9].

Proposed Workflow

1. Pre-Procedure

- Model loads SSS protocol requirements (10 images for upper GI).

2. Live Guidance

- Flags missed zones (e.g., "Antrum not fully visualized – adjust scope angulation").

3. Post-Procedure

- Generates heatmaps of examined regions and confidence scores for pathology detected.

Impact: Reduces missed lesions, standardizes exams across skill levels, and provides auditable documentation – addressing gaps highlighted in^{[7] [1] [4]}.

Conclusion

Your model aligns with ESGE/ASGE guidelines^{[7] [3]} and evidence on AI's role in reducing diagnostic variability^{[5] [4]}. By combining anatomic landmark detection with bias-aware algorithms, it addresses both technical and cognitive limitations in current practice.

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1. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9157695/>
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